

Performance Evaluation of Lightweight Classifiers for Early-Stage Parkinson’s Disease Detection using Acoustic Features

Abstract—Parkinson’s disease is a progressive neurodegenerative disorder, and early diagnosis is critical. However, traditional methods are subjective, and current deep-learning model-based solutions require large computational power, which is not suitable for everyday medical use. This study evaluated the performance of lightweight machine learning classifiers for early-stage Parkinson’s detection using acoustic (voice) features. Seven different lightweight machine learning models were evaluated on the Oxford Parkinson’s Disease Detection Dataset (sourced from UCI) using Z-score normalization, a 5-fold GroupKFold cross-validation approach to prevent subject-level data leakage, and SMOTE (Synthetic Minority Over-Sampling Technique) to manage class imbalance. After the evaluation, the XGBoost model was identified as the lead model by achieving a mean cross-validation F1-score of 0.900 and a final recall of 0.97 on unseen test subjects. While the model demonstrated a high false-positive rate due to the dense overlap of acoustic features between healthy and Parkinson’s instances, the high recall ensures that the model rarely misses a positive case. These findings demonstrate that XGBoost is the most effective lightweight classifier for this task, even though its overall performance dropped on the test data due to the inherent problems within the dataset.

Index Terms—Parkinson’s Disease, Machine Learning, XGBoost, Acoustic Features, Early Detection, Lightweight Classifiers

I. INTRODUCTION

Parkinson’s Disease (PD) is a progressive neurodegenerative disorder that results in total or partial damage to the voice, behavior, and other necessary skills, making early detection critical for managing symptoms [1]. In the past, clinical diagnoses heavily depended on physical neurological exams. However, these traditional methods are time-consuming and expensive, and many misdiagnoses happen because patients entirely depend on the subjective assessment of the physician [2]. As a solution, recent medical diagnosis has shifted towards non-invasive biomarkers. Because vocal impairment (dysphonia) is one of the earliest quantifiable indicators of Parkinson’s, analyzing acoustic features provides a fast, objective, and highly accessible way to detect the disease [3].

To improve the accuracy of early diagnosis of Parkinson’s disease, machine learning (ML) and deep learning (DL) techniques are now used frequently. However, a major limitation in current methods is the reliance on highly complex models such as massive neural networks. These complex models require massive computational power, making them difficult to implement in simple, everyday medical environments. Also, recent studies show that better diagnostic accuracy and efficiency can be acquired if the complexity of these traditional complex

methods is reduced. Therefore, the demand for lightweight machine learning classifiers that are capable of accurately detecting Parkinson’s disease from voice signals and simple deployment in practical healthcare settings is increasing [4].

The primary aim of this study is to evaluate the performance of seven lightweight machine learning classifiers for early-stage Parkinson’s detection using acoustic features and find the optimal model among them. To achieve this, a systematic evaluation was conducted on these seven different models using the Oxford Parkinson’s Disease Detection Dataset sourced from the UCI Machine Learning Repository [5]. This procedure included a strict GroupShuffleSplit to prevent subject-level data leakage, standard scaling, and 5-fold cross-validation to ensure the reliability and clinical validity of results. As the final outcome, the XGBoost model was identified as the lead model among these seven models. The XGBoost model proved that computationally simple algorithms can achieve high-sensitivity screening performance by achieving a mean cross-validation F1-score of 0.900 and a final recall value of 0.97 on unseen test subjects.

II. RELATED WORK

The early groundwork for using voice analysis in Parkinson’s detection was laid by Little et al. [3], [6], who demonstrated that nonlinear acoustic measures can capture subtle vocal changes associated with the disease. More recent studies have explored deep learning approaches, such as the CNN-based framework proposed by Saha and Nath [4], which achieve strong performance but depend on computationally heavy architectures. These models are difficult to deploy in routine clinical settings, where resources are limited. Because of this gap, the present study focuses on a direct comparison of seven lightweight machine learning classifiers. Using established acoustic features, the goal is to determine whether simpler models can provide reliable diagnostic performance without the complexity and hardware demands of deep learning systems.

III. METHODOLOGY

A. Dataset Acquisition

The Oxford Parkinson’s Disease Detection Dataset was acquired from the UCI Machine Learning Repository to train and evaluate the models. This dataset consists of 195 voice recording instances, with each instance containing 22 acoustic features.

The dataset contains 48 healthy instances and 147 Parkinson's instances, resulting in a significant class imbalance of approximately a 3:1 ratio. This class imbalance must be addressed in order to prevent machine learning models from developing a predictive bias toward the majority class.

Crucially, these 195 instances are not derived from 195 independent people; rather, they are derived from a localized group of only 31 distinct patients (averaging approximately six vocal recordings per patient). In standard machine learning workflows, non-predictive columns such as subject identifiers are removed from the dataset. However, applying that standard practice to this dataset introduces a severe risk of subject-level data leakage. To ensure a comprehensive and valid clinical evaluation, the subject identifiers (patient IDs) were explicitly extracted and retained in a separate location before the data preprocessing began. This ensured that individual patients could be safely isolated during the subsequent data-splitting process.

B. Acoustic Feature Characterization

Although the primary data source for these 22 acoustic features is human vocals, this research does not process raw analog audio signals. Instead, the dataset consists of extracted numerical descriptors that quantify specific physical properties of the voice [3]. Each individual feature represents a statistical or mathematical summary of a 30-second sustained vowel phonation. This transformation from analog data to digital numerical data allows lightweight classifiers to perform high-level analysis without the computational overhead of a dedicated signal-processing layer.

To ensure the effectiveness of the classifiers in distinguishing between healthy and Parkinson's states, it is crucial to understand the biomechanical significance of these numerical acoustic features. Extracted from vowel phonations, they are categorized into five primary groups based on their signal processing characteristics [3], [5]:

- **Fundamental Frequency (F0) Metrics:** Features including MDVP:F0(Hz), MDVP:Fhi(Hz), and MDVP:Flo(Hz) capture the average, maximum, and minimum pitch of the voice. In early-stage Parkinson's, laryngeal muscle rigidity often results in reduced frequency ranges, which manifests clinically as a 'monotone' vocal output [3].
- **Vocal Micro-perturbation (Jitter and Shimmer):**
 - *Jitter*: Includes features such as MDVP:Jitter(%) and Jitter:DDP, which represent the cycle-to-cycle variation in fundamental frequency. This quantifies vocal instability and a 'shaky' voice, a strong early symptom of Parkinson's [3].
 - *Shimmer*: Represented by features like MDVP:Shimmer and MDVP:APQ, measuring the variation in amplitude (loudness) of the signal. High shimmer levels are clinically associated with breathiness and hoarseness [3].
- **Noise and Harmonic Ratios:** The Harmonics-to-Noise Ratio (HNR) and Noise-to-Harmonics Ratio (NHR) measure the clarity of the voice. A high NHR level indicates

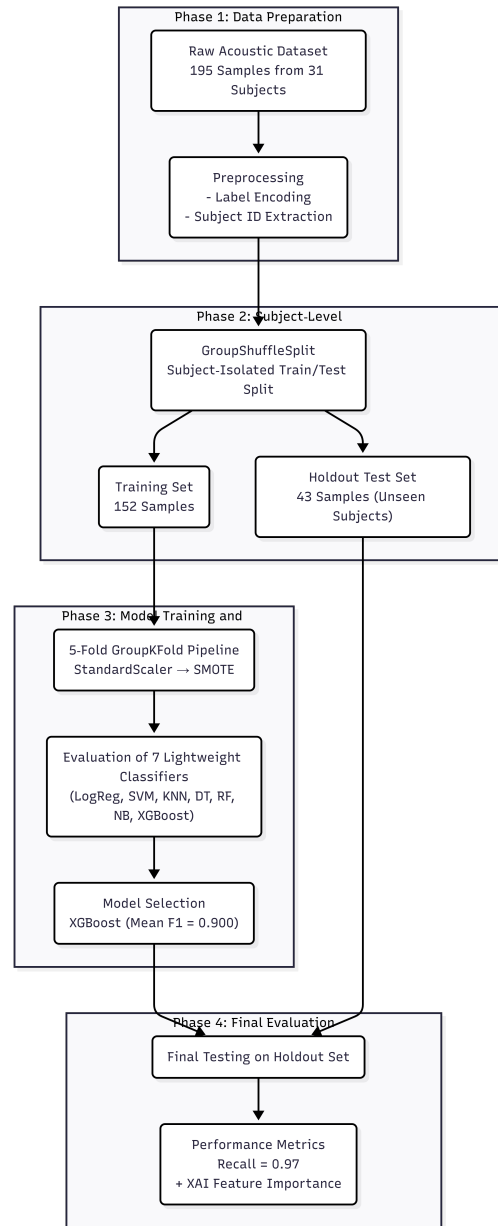


Fig. 1. Proposed system architecture for early-stage Parkinson's disease detection.

increased turbulence in the airflow, indicating that the vocal folds are not completely closing during phonation [3].

- **Nonlinear Complexity Measures:** Features like RPDE (Recurrence Period Density Entropy) and D2 (Correlation Dimension) quantify the chaos and complexity within the vocal signal [6]. Unlike linear measures, these capture the unpredictable state of voice degradation that occurs as the nervous system loses fine motor control.
- **Pitch Period Entropy (PPE) and Spread:** PPE, spread1, and spread2 are advanced measures specifically designed to be sensitive to the speech changes unique to Parkinson's. PPE demonstrates strong robustness to the natural

variations found in healthy vocal production, making it an essential feature for detection [3].

C. Subject-Aware Data Partitioning

Prior to feature scaling and augmentation, the dataset required partitioning. Because multiple vocal recordings belong to the same patient, applying a standard randomized train-test split would randomly scatter individual patient recordings across both the training and testing sets. This phenomenon, known as subject-level data leakage, allows a model to achieve artificially high accuracy simply by memorizing a specific patient's acoustic signature rather than learning generalized disease markers.

To prevent this leakage, the isolated patient IDs were utilized to perform a strict GroupShuffleSplit. Instead of splitting individual recordings, the dataset was partitioned strictly by patient IDs. This method isolated a 20% holdout testing set consisting of 43 instances. Because of this grouping mechanism, every vocal instance in the holdout set originated from patients that the model had never encountered during training, ensuring that the final evaluation reflects genuine clinical generalization.

D. Data Preprocessing and SMOTE Augmentation

In this dataset, the default acoustic features are measured in different units. This causes certain features to have larger numerical ranges than others, which can bias the models and make smaller features, such as Jitter and Shimmer, mathematically "invisible" to the machine learning algorithms.

To solve this problem, standardization (Z-score normalization) was used to transform the acoustic features to a mean (μ) of 0 and a standard deviation (σ) of 1. This ensured that all acoustic features were represented on a single, uniform mathematical scale. Each feature was transformed using the following equation:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Where:

- z : The resulting standardized score (Z-score).
- x : The original raw feature value (e.g., a specific Hertz value).
- μ : The mean of the feature in the training set.
- σ : The standard deviation of the feature in the training set.

Also, to fix the class imbalance caused by 147 Parkinson's and 48 healthy instances, a Synthetic Minority Over-Sampling Technique (SMOTE) was used. This generates mathematically realistic, synthetic instances of the minority (healthy) class to perfectly balance the training set.

E. Classifier Selection and Pipeline Construction

To find the most effective approach for Parkinson's detection, seven lightweight machine learning models were evaluated instead of using deep learning models, which require comparatively higher computational power. The evaluated

models include different types of algorithms to determine which mathematical approach works best for acoustic data:

- **Linear Models:** Logistic Regression
- **Distance-Based Models:** Support Vector Machine (SVM) and K-Nearest Neighbors (KNN)
- **Tree-Based Models:** Decision Tree and Random Forest
- **Boosting Models:** Extreme Gradient Boosting (XGBoost)
- **Probabilistic Models:** Naive Bayes

For the standard models, the scikit-learn library was used, and the official XGBoost library was used for the XGBoost model. To completely prevent data leakage, the scaling, SMOTE augmentation, and training phases were strictly packaged into an execution pipeline (Standard Scaler → SMOTE → Classifier).

F. Evaluation Metrics and Validation

To evaluate the performance of the models, the dataset was split into 80% training and 20% testing data. To ensure rigorous evaluation and prevent overfitting, a 5-fold GroupKFold cross-validation approach was applied to the training data to ensure no patient overlap occurs between the training and validation folds.

Because of the class imbalance in this dataset, depending only on the accuracy is not effective. Therefore, the models were evaluated using accuracy, precision, recall, ROC-AUC, and F1-score. The F1-score is important for this research because it provides a balance between precision and recall, ensuring models can identify Parkinson's patients without being biased by the class imbalance.

G. Explainable AI (XAI) Biomarker Extraction

A major barrier to the adoption of machine learning in clinical settings is the "black box" nature of complex algorithms. To provide transparency, an Explainable AI (XAI) approach was used to extract the feature importance matrix from the champion algorithm. This allows the study to rank which specific acoustic features are the primary drivers of Parkinson's detection.

IV. RESULTS AND DISCUSSION

A. Performance Comparison and Cross-Validation

All seven classifiers' performance was evaluated using a 5-fold GroupKFold cross-validation approach. As shown in Table I, XGBoost is identified as the best model of all 7 classifiers, making an F1-score of 0.900 and an accuracy of 0.849. Random Forest and SVM also showed excellent performance. Also, Naive Bayes had the lowest predictive accuracy, which is 0.704.

B. Final Model Evaluation and Error Analysis

To evaluate the final performance of the XGBoost model, which had the highest accuracy and F1-score, the 20% holdout testing data were used. That 20% test data contained 43 total instances, including 31 Parkinson's and 12 healthy instances.

TABLE I
5-FOLD CROSS-VALIDATION PERFORMANCE OF SEVEN LIGHTWEIGHT
CLASSIFIERS ON THE PARKINSON’S VOICE DATASET

Model	Accuracy	F1-Score	Recall	ROC-AUC
XGBoost	0.849	0.900	0.921	0.861
Random Forest	0.797	0.867	0.894	0.835
SVM	0.803	0.866	0.863	0.826
Decision Tree	0.777	0.855	0.879	0.656
KNN	0.783	0.837	0.792	0.847
Logistic Regression	0.770	0.843	0.815	0.867
Naive Bayes	0.704	0.782	0.729	0.810

From the final evaluation, XGBoost achieved an overall accuracy of 0.721 (or 72.1%) and a target F1-score of 0.833 for the Parkinson’s class. Furthermore, a recall of 0.97 shows that the XGBoost model rarely misses a Parkinson’s patient and has good generalization capabilities.

As shown in Fig. 2, the confusion matrix analysis reveals that while the XGBoost model correctly identified 30 Parkinson’s patients, it only correctly identified 1 healthy patient. Furthermore, 11 healthy patients were falsely flagged as having Parkinson’s, and only 1 Parkinson’s patient was incorrectly flagged as healthy. Although the model achieved a near-perfect recall by missing only 1 out of 31 sick patients, this cannot be justified as a clinically viable outcome. The misclassification of nearly all healthy patients (11 out of 12) indicates a severe predictive bias toward the Parkinson’s class. The application of SMOTE during training caused the algorithm to overcompensate, resulting in a model that prioritizes Parkinson’s detection at the extreme expense of specificity. This highlights a critical limitation: the acoustic features of healthy and Parkinson’s patients in this dataset overlap too heavily for lightweight classifiers to separate them without generating an unacceptably high rate of false positives.

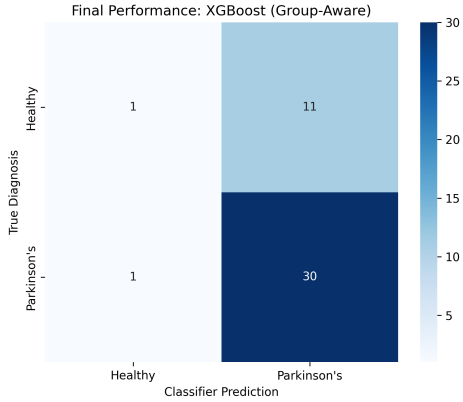


Fig. 2. Confusion matrix of the XGBoost classifier evaluated on the 20% holdout testing set, illustrating the high false-positive rate.

C. Diagnostic Evaluation and Biomarker Extraction

A Receiver Operating Characteristic (ROC) curve was used to assess overall diagnostic performance, resulting in an Area Under the Curve (AUC) of 0.599 (Fig. 3). Although any

value exceeding 0.5 confirms the model performs better than random classification, this specific score directly points to the previously noted predictive bias and elevated false-positive rate. The shape of the curve confirms that while the algorithm easily detects the minority class, the dense acoustic overlap between Parkinson’s and healthy subjects prevents it from effectively balancing sensitivity with specificity.

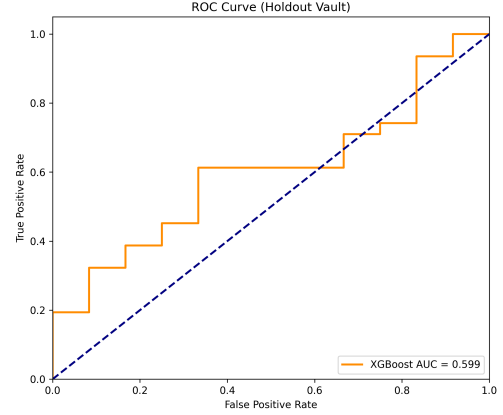


Fig. 3. Receiver Operating Characteristic (ROC) curve of the XGBoost classifier, yielding an AUC of 0.599 and illustrating the challenge of balancing sensitivity with specificity.

An Explainable AI (XAI) approach was used to extract feature importances directly from the XGBoost model (Fig. 4). The analysis indicated that MDVP:Jitter(Abs) is overwhelmingly the most critical vocal biomarker, accounting for nearly 48% of the model’s decision-making weight. This confirms clinical literature, as absolute jitter measures the cycle-to-cycle variation in vocal fundamental frequency, highlighting that micro-tremors and vocal instability are main symptoms of Parkinson’s disease. Other secondary features included spread2 (~12.5%), the fundamental frequency MDVP:F0(Hz) (~9.3%), and spread1 (~9.3%). These biomarkers provide clinical transparency, allowing medical professionals to analyze the acoustic measurements.

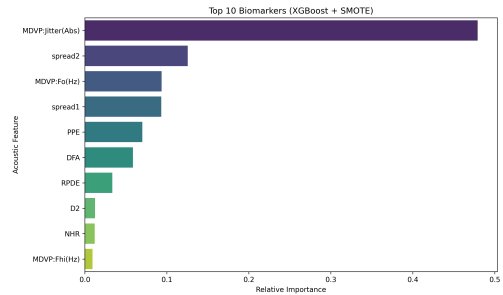


Fig. 4. Feature importance ranking extracted from the XGBoost model, identifying MDVP:Jitter(Abs) as the primary acoustic biomarker for detection.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

In summary, the identification of XGBoost as the best-performing lightweight classifier proves that the primary goal of this research was addressed. During the 5-fold GroupK-Fold cross-validation phase, XGBoost outperformed all other models with a mean F1-score of 0.900. When subjected to the final evaluation on the unseen holdout testing set, the model achieved a final accuracy of 0.721 and a remarkable recall of 0.97.

However, the evaluation on unseen patients revealed that while the model is highly sensitive to Parkinson's symptoms, it maintains a high false-positive rate. Therefore, this model is currently most effective as a high-sensitivity, first-line clinical screening tool rather than a standalone diagnostic system.

B. Future Work

Despite these results, several paths remain open for future investigations. While the Oxford Parkinson's Disease Detection Dataset is a very high-quality dataset, having only 195 instances from only 31 distinct patients might not be enough to train a model that can identify globally diverse acoustic voices and languages. With a larger global dataset, this problem can be addressed.

Furthermore, future studies can be expanded to other types of biological data like handwriting or gait analysis alongside acoustic features to make the model even more accurate and reduce false positives.

To address this false-positive bias, future studies should move toward multimodal data integration to achieve higher diagnostic precision. Furthermore, exploring deep learning architectures, such as Convolutional Neural Networks (CNNs) on voice spectrograms, may lead to identifying hidden patterns that standard numerical features overlook.

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